14 The reaction of hydrogen peroxide with iodide ions in an acidic solution can be monitored by an initial rates method.

$$H_2O_2(aq) + 2H^+(aq) + 2I^-(aq) \rightarrow 2H_2O(l) + I_2(aq)$$

The rate equation was found to be as follows:

$$\text{rate} = k\,[H_2O_2][I^-]$$

What of the following mechanism correctly describes this reaction?

A $H_2O_2 + I^- \rightarrow H_2O + IO^-$ (slow)
$IO^- + H^+ \rightarrow HIO$ (fast)
$HIO + H^+ + I^- \rightarrow H_2O + I_2$ (fast)

B $H_2O_2 + I^- \rightarrow H_2O + IO^-$ (slow)
$H_2O_2 + IO^- \rightarrow H_2O + IO_2^-$ (fast)
$IO_2^- + I^- + 4H^+ \rightarrow 2H_2O + I_2$ (fast)

C $2H^+ + 2I^- \rightarrow 2HI$ (fast)
$2HI + H_2O_2 \rightarrow 2H_2O + I_2$ (slow)

D $H_2O_2 + I^- + H^+ \rightarrow H_2O + HIO$ (fast)
$HIO + I^- \rightarrow OH^- + I_2$ (slow)
$OH^- + H^+ \rightarrow H_2O$ (fast)